

Koundinya N. V. S. S.

✉ koundinya.nvss@gmail.com in koundinya-nvss

Summary

AI Architect with 7+ years of experience leading the design and delivery of production-grade AI systems across healthcare, travel technology, public sector, and insurance domains. Founding team member at a healthcare AI startup, driving architecture decisions for a platform targeting a billion users. Proven ability to build and lead cross-functional teams, define organization-wide AI strategy, secure executive buy-in through C-suite and investor presentations, and translate complex technical architectures into business outcomes.

Experience

Mondee Tech Private Limited — AI Architect | Founding Team

May 2025 – Present

Stealth-mode Healthcare AI startup | Three product lines targeting Indian consumers | Concurrent consulting for travel tech, public sector, and insurance enterprise clients

- Serve as the **AI Architect** with full ownership of architecture, safety engineering, compliance, and developer enablement — with authority to make and defend framework-level decisions in front of the CEO, investors, and C-suite executives.
- Defined the company's **framework-agnostic microservices** principle as a non-negotiable architectural constraint, ensuring durability as the AI landscape evolves.
- Translated complex architectures into executive presentations and investor-ready business narratives using **architecture diagrams**, and **metric-driven slides** — securing executive approval for strategic initiatives including cross-lane medical coding integration.
- Authored the **alpha release resource and risk plan**, identified critical under-resourcing (115% utilisation) and zero QA coverage, and escalated findings to leadership with a structured mitigation plan.
- Designed an **8–10 week Developer Acceleration Program** and a **3-week AI/ML upskilling program** (100+ employees, 4 tracks), authored CTO-commissioned technical playbooks, and established experiment tracking systems for cross-company knowledge sharing.
- Drove active client engagements for a state law enforcement agency (primary AI architecture lead) and a leading insurance enterprise through Gen AI strategy presentations mapping 20+ use cases across the value chain

Digi Quanta — AI Solutions Architect

Feb. 2025 – Apr. 2025

Acqui-hired by Mondee in April 2025 | 3 clients across media/entertainment, healthcare AI, and US home health

- Led a cross-functional team of 11, running agile sprints and managing concurrent delivery timelines across 8 AI product initiatives spanning speech, vision, NLP, and predictive ML.
- Established team-wide best practices: standardized client-facing artifact delivery framework (architecture docs, model documentation, compliance, UAT, handover), project folder structure, and reusable project documentation templates.
- Responsible for solutioning, architecture design, implementation oversight, and stakeholder-facing demos across all engagements.

Genpact India (Positions held: **Manager**, Assistant Manager, Business Analyst and Data Science Intern)

Jul. 2019 – Jan. 2025

Data science leadership across healthcare, insurance, financial services, and e-commerce

- Worked closely with stakeholders to translate business requirements into actionable insights, personalize solutions to boost user adoption, and deliver real-time demo applications that accelerated project delivery and executive buy-in.
- Managed a team of data scientists and mentored 10+ interns through structured programs, providing technical guidance on cutting-edge ML techniques, research paper writing, and career development — multiple interns converted to full-time hires.
- Aligned team objectives with business goals through performance management frameworks (GitHub Projects), drove process improvements through automation, and collaborated cross-functionally with UI/UX, data engineering, and stakeholder teams.
- Led knowledge-sharing culture: authored 16 peer-reviewed research papers, filed 1 patent, and conducted organization-wide technical sessions on emerging AI techniques.

Additional Experience

AI for Social Impact (Swecha NGO): AIML Core Team, Tech Speaker & Workshop Facilitator

- Organizing Committee Member for AI Days 2025; led end-to-end management of the Spark Camp program.
- Conducted a 2-day workshop “From Concept to Code: A Developer’s Guide to Agentic Frameworks” at AI Days 2024, Hyderabad.
- Delivered talks on Prompt Engineering, AI Essentials, Continuous Integration. Organized workshops on LLM Fine-tuning and Low Code/No Code AI. Contributed to faculty development programs on AI

President - Public Policy Club, BITS Pilani

- Led club growth and secured official recognition from the BITS Pilani Student Union.

Event Coordinator - Andhra Samithi, BITS Pilani

- Managed 50-person team for the successful execution of the Yuva cultural night event.

Education

Birla Institute of Technology and Science, Pilani CGPA: 7.38 | B.E. Mechanical Engineering & M.Sc. Mathematics

2013 – 2018

Skills

Areas of Expertise: Agentic AI, Virtual Avatars, Multi-Agent Orchestration, AI Safety & Guardrails, Context Engineering, Generative AI, Machine Learning, Computer Vision, Natural Language Processing, Reinforcement Learning, Data Analytics & Visualization

Cloud Platforms and Tools: AWS, Azure, GCP, Dify, LangFuse, n8n, Dataiku, HyperScience, FlowiseAI, ComfyUI

Libraries/Frameworks: LangChain, LangGraph, Google ADK, LiveKit, FastMCP, Transformers, Unsloth, Scikit-learn, sktime, Flask, FastAPI, Gradio

Projects

Multi-Agent Avatar based Conversational Health Assistant Platform (Sub-2s, 7 Specialties)

- Architected a unified multi-agent system covering 7 medical specialties. Designed Registry-Strategy-Factory-Builder dispatch pattern; implemented consent based agent consultation with automatic EHR generation.
- Designed guardrail microservice (PII/injection/jailbreak screening + hallucination output filtering) fully decoupled from any agent SDK

Technologies Used: Python, LiveKit, Qdrant, FastMCP, Mem0, SQLite3, sqlite-vec, FastAPI, Cartesia, Soniox, LangFuse

Clinical Decision Support System (CDSS) and Clinical Knowledge Bases

- Developed a voice-first medical AI achieving 82% word accuracy and 87% clinical response accuracy
- Designed a clinical reasoning Knowledge Graph using SNOMED CT to infuse determinism and predictability into LLM responses by grounding them in these authoritative knowledge bases.

Technologies Used: Python, LangGraph, LiveKit, memgraph, FastMCP, SQLite3

Multilingual FIR Automation Application

- Architected a full-stack platform for automated complaint intake for a state law enforcement agency; evaluated 5 ASR providers across 8 parameters; designed hybrid deployment scaling from 50 to 900 devices.

Technologies Used: Python, ASR systems, MongoDB, Flutter

Medical Document Processing & Re-Hospitalization Risk Profiling

- Built AI pipeline for document classification, ICD code matching via RAG, and YOLO-based signature detection. Separately built an ML pipeline (Random Forest, LightGBM, XGBoost) predicting re-hospitalization risk from OASIS data.

Technologies Used: Python, YOLO, RAG, HuggingFace Transformers, LightGBM, XGBoost, Scikit-learn

Voice Cloning & Lip Sync Pipelines

- Guided in the development of RVC Voice Cloning (4.3/5 similarity from 3–5 min audio) and Lip Sync pipelines (4.5+/5 alignment, sub-300 ms latency), benchmarked favorably against Adobe Character Animator.

Technologies Used: Python, RVC, PyTorch, LatentSync, Whisper, OpenCV, Dlib, CUDA

Intelligent Research Assistant: Q&A Chatbot with RAG Fusion

- Deployed a RAG Fusion approach outperforming baseline RAG perplexity by 15% for accurate research responses with cross-references and integrated RAGAS metrics for real-time confidence scoring

Technologies Used: Python, Azure OpenAI, Azure Search, Azure functions

Multilingual Medical Invoice Processing using Multimodals

- Developed a multimodal-based document translation and extraction pipeline achieving 72% accuracy, 2-minute response times (down from 2 days), and substantial cost savings.

Technologies Used: Python, Azure OpenAI (GPT4-Vision), Azure functions

GenAI-driven Content Mining from Emails

- Developed a zero-shot email order extraction pipeline with 86% accuracy using prompt engineering and API optimizations.

Technologies Used: Python, Langchain, Azure OpenAI (GPT4-Vision), Azure functions, Azure Storage

Machine-Learning powered SG&A Cost Forecasting for Strategic Financial Planning

- Delivered SG&A expense forecasts (MAPE < 0.1) via univariate models with exogenous factors; streamlined pipelines to boost financial reporting efficiency by 30%.

Machine Learning Powered Reinsurance Risk Assessment

- Led data analysis and stakeholder collaboration to develop an ML-powered claims queue system with business rules, boosting revenue 15% and cutting underwriter review time 40%.

Sales Order Hold Analytics; Predictive Modelling for Financial Planning and Analysis

- Analyzed order-to-cash processes and delivered ETL jobs, statistical/ML forecasting models, and recommendations optimizing cash flow (\$0.5M impact) plus insightful earnings guidance for a multinational healthcare conglomerate.

AI Automated Automotive Insurance Claims

- Improved the mIoU of parts detection models from 52% to 68% using detectron-2, which minimized parts misclassification and enhanced the overall pipeline accuracy by ~12%.

Automated Nutrition Label Reforms

- Developed a solution to reduce human efforts and time by almost 95% through an automated verification process for nutrition labels by employing a combination of CV, NLP, and custom-built recursive partitioning algorithms.

Crop Yield Predictions with Deep Learning

- Built a deep learning solution, involving Convolutional Neural Networks, to predict crop yield using multi-spectral satellite imagery taking cues from the paper [Deep Gaussian Process for Crop Yield Prediction Based on Remote Sensing Data](#).

Patents and Publications

• Dimension Estimation Using Duplicate Instance Identification In A Multiview And Multiscale System	2024
Patent Link	
• Video RAG: Leveraging Semantic Chunking and Retrieval Augmented Generation for Enhanced Information Extraction - ICECET 2025	2025
10.1109/icecet63943.2025.11471900	
• Evaluating Modern Information Extraction Techniques for Complex Document Structures - ICECET 2024	2024
10.1109/ICECET61485.2024.10698618	
• Harnessing Multimodal AI for Creative Design - Performance Evaluation of Stable Diffusion and DALL-E 3 in Fashion Apparel and Typography – ICECET 2024 Space@123	2024
10.1109/ICECET61485.2024.10698398	
• A Deep Dive into RAG Paradigms: Selecting the Right Approach for Specific Business Applications – 11th ICBAI, IIM Bengaluru	2024
• Transforming Video Content into Dynamic Multimodal SOPs – 11th ICBAI, IIM Bengaluru	2024
• Advancing Engineering Drawing Interpretation: A Comparative Study of Prompting Techniques – 11th ICBAI, IIM Bengaluru	2024
• Optimizing Multimodal AI Performance: A Comparative Analysis of Prompting Techniques Across GPT-4O, Gemini, and Claude – 11th ICBAI, IIM Bengaluru	2024
• Comparative Evaluation of Advanced AI Models and Services for Complex Tabular Data Extraction from Documents – 11th ICBAI, IIM Bengaluru	2024
• Analyzing and Mitigating Prompt Injection Techniques in AI Language Models – 11th ICBAI, IIM Bengaluru	2024
• Benchmarking Advanced Video Generation Models: Evaluating Effectiveness in Creative Media and Corporate Communication – 11th ICBAI, IIM Bengaluru	2024
• Revolutionizing Contract Management with Generative AI: Uncovering Loopholes with Precision and Speed – 11th ICBAI, IIM Bengaluru	2024
• Automated Content Moderation Using AI: An Agentic Approach to Process Text, Audio, Video, and Images – 11th ICBAI, IIM Bengaluru	2024
• A Novel Approach for Evaluating Performance Consistency in Time Series Forecasting Models – 11th ICBAI, IIM Bengaluru	2024
• ML Powered Life Insurance Risk Selection Leading to Efficient and Streamlined Underwriting Process – 7th ICBAI, IIM Bengaluru	2022
• Statistical Modeling of Solar Energy chapter of Enhancing Future Skills and Entrepreneurship book SPLCEM Springer series	2020
10.1007/978-3-030-44248-4_16	
• A Novel Machine Learning Based Approach for Generic Form Processing – 5th ICBAI, IIM Bengaluru	2017